Cochrane RoB-assessment tool for randomised trials [Sterne et al. (2019); Higgins et al. (2023)]
See link for guidance

Dias et al. (2015)				**COMPLETE**
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?	Y			"subjects were randomly assigned by drawing lots in two groups"
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	No information is provided regarding who generated the random sequence.
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N			"There were no significant differences between groups for age (p = 0.433), body mass (p=0.615), or height (p=0.515)."
Risk-of-bias judgment (low/high/some concerns)	Some concerns			
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		PY		It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY			No mention of outside training, but since they are untrained, probably no training was performed.
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None mentioned, but seem unlikely
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN			Some dropped out, as is expected, but a similar number from each group "seven volunteers dropped out for personal reasons (i.e., four from ETG and three from CTG). "
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?	Y			"A two-way ANOVA [group (ETG × CTG) vs. time (base- line vs. post-6 vs. post-12)], followed by a LSD post hoc test, was used to compare absolute values of leg press 1RM, knee extension 1RM"
Risk-of-bias judgment (low/high/some concerns)	Low risk			
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?		N		≈27% of randomised participants dropped out for personal reasons.
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?		PN		No sensitivity analyses. Only the complete cases are of interest

3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	PN		I see no reason to expect that dropouts were related to the outcome. Personal reasons were reported
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Assessments and time of asssessment were similar in bopth groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	Blinding is not mentioned
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	PY		Yes, more encouragement could be given. Or attempts could be added.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		I don't see any reason to suspect the outcome was influenced. There's no obvious bias for either treatment and there was a null effect reported.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in selection of the reported result			
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:			
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?		NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?		NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?			
Overall bias			
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?			
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.			
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*			

Pereira et al. (2016)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	They only mentioned that they were randomized, nothing else
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	They only mentioned that they were randomized, nothing else
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN			Groups looks very similar at baseline, both for age, height, body mass, MT and RM.
Risk-of-bias judgment (low/high/some concerns)	Some concerns			
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		PY		It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?			NI	No information regarding outside training
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN			All subjects recruited seemed to finish.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?	Y			"ANOVA for repeated measured was used to verify possible differences in strength and hypertrophy gains between time and groups. Significance P-level ≤0.05 was accepted. When the F test was significant, complement analysis by Bonferroni's multiple comparison tests was made."
Risk-of-bias judgment (low/high/some concerns)	Some concerns			
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y			The study had no dropouts, and all measurements are reported
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?				
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?				
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?				

Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	PY		Yes, more attempts or more encouragement could be given if there is a bias for one intervention over another.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PY		It is reasonable to suspect that the outcome assessors were not blind to the intervention received, this could indicate a bias that may affect assessment of the outcome. The conclusions of the study (which used within-condition comparisons to make a between condition conclusion that longer ECC is better than shorter ECC) suggest there is plausibly a conscious or subconscious bias towards longer eccentric tempos.
Risk-of-bias judgment (low/high/some concerns)	High risk		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - HYPERTROPHY			
4.1 Was the method of measuring the outcome inappropriate?	PN		Ultrasound, although more prone to human error and judgement than MRI and CT, has been found valid as a measure of hypertrophy.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	PY		Given that ultrasound is inherently prone to human error and judgement, it could give rise to measurement error.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Given the subject and the neutral initial objective of the study (exploratory), I find it unlikely.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in selection of the reported result			
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols

Is the numerical result being assessed likely to have been selected, on the basis of the results, from:				
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?			NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?			NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?				
Overall bias				
Risk-of-bias judgment (low/high/some concerns)		High risk		For both, high risk domain for strength, multiple "some concerns" for hypertrophy that we judge substantially lower confidence in the results.
Optional: What is the overall predicted direction of bias for this outcome?				
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.				
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Mike et al. (2017)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	It just mentions randomisation, but not how.
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	Same as above
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN			Baseline data were similar for anthropometrics and strength.
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		PY		It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY			No other lower-body training was allowed outside training
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent

2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN		Some dropped out, as is expected, but groups had a similar number of participants at the end, so it unlikely to have affected results.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to deviations from intended interventions?			
Bias due to missing outcome data			
3.1 Were data for this outcome available for all, or nearly all, participants randomised?		N	25% of randomised participants dropped out due to adherence
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?		NI	No sensitivity analyses. Only the complete cases are of interest
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	PN		I see no reason to expect that dropouts were related to the outcome. Adherence was the reason and it was equally distributed between groups-
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		PY	No mention of blinding, it seems likely that those delivering the intervention were also at testing
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	Yes, more encouragement, more attempts, etc.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		It seems outcome assessors were not blind to the intervention received, this could indicate a bias that may affect assessment of the outcome. However, I would not judge this to be likely given the null effects reported.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - POWER			
4.1 Was the method of measuring the outcome inappropriate?	N		Vertical jump reach height via vertec is a valid measure of jump height

4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN			Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?			NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY		Yes, more encouragement, more attempts, etc.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN			It would seem unlikely that belief about longer eccentric contractions could have biased the results in favour of this outcome.
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?				
Bias in selection of the reported result				
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?			NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:				
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?			NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?			NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?				
Overall bias				
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?				
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.				
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Kojic et al. (2021)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	No mention of methods, just randomised.
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	Same as above

1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN		Baseline looked similar.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias arising from the randomisation process?			
Bias due to deviations from intended interventions			
2.1 Were participants aware of their assigned intervention during the trial?		PY	It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY	No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY		No mention of outside training, but since they are untrained, probably no training was performed.
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN		None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN		Some dropped out, as is expected, but groups had a similar number of participants at the end, so it unlikely to have affected results.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to deviations from intended interventions?			
Bias due to missing outcome data			
3.1 Were data for this outcome available for all, or nearly all, participants randomised?		N	≈17% of randomised participants dropped out due to adherence
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?		NI	No sensitivity analyses. Only the complete cases are of interest
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	PN		I see no reason to expect that dropouts were related to the outcome. Adherence was the reason and it was equally distributed between groups-
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.

4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	Additional encouragement or attempts are possible.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Our judgement is that the researchers have no strong level of belief in favour of either intervention.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - HYPERTROPHY			
4.1 Was the method of measuring the outcome inappropriate?	PN		Ultrasound, although more prone to human error and judgement than MRI and CT, has been found valid as a measure of hypertrophy.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	Given that ultrasound is inherently prone to human error and judgement, it could give rise to measurement error.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Given that it seems an independent researcher took the ultrasound, I don't see much of a reason to suspect the outcome was substantially influenced by the intervention received.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in selection of the reported result			
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:			
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?		NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?		NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?			
Overall bias			
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?			

Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.				
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Shibata et al. (2021)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	No mention of methods, just randomised.
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	Same as above
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN			Baseline looked similar.
Risk-of-bias judgment (low/high/some concerns)	Some concerns			
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		PY		It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY			There is an attempt to control nutrition but I see no information about controlling for training outside of the intervention.
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN			Some dropped out, as is expected, but groups had a similar number of participants at the end, so it unlikely to have affected results.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?				
Risk-of-bias judgment (low/high/some concerns)	Low risk			
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?		N		≈8% of randomised participants dropped out due to adherence
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?			NI	No sensitivity analyses. Only the complete cases are of interest

3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	PN		I see no reason to expect that dropouts were related to the outcome. Injury and illness are the reasons.
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		PY	It seems reasonable to suspect that the researchers delivering the intervention were also the outcome assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	More encouragement, attempts etc.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Given the neutral hypothesis and the result favoring short ecc tempo, I don't think it is likely the outcome was influenced by knowledge of the intervention.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - HYPERTROPHY			
4.1 Was the method of measuring the outcome inappropriate?	N		MRI is considered gold standard for hypertrophy
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No information on who conducted the MRI.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	PN		There is some effort described (i.e., image shuffling) that indicates the researchers were aware of potential bias so I just this as not likely to have occurred.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - POWER			
4.1 Was the method of measuring the outcome inappropriate?	N		CMJ through camera seems appropriate
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.

4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PN		There is less judgement in this measure since frames during flight is the outcome unit.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?				
Risk-of-bias judgment (low/high/some concerns)		Low risk		
Optional: What is the predicted direction of bias in measurement of the outcome?				
Bias in selection of the reported result				
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?			NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:				
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?			NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?			NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?				
Overall bias				
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?				
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.				
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Pearson et al. (2022)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?	PY			"After subjects performed the 1RM back squat to meet the inclusion criteria, their legs were stratified into terciles based on the sum of MT (SMT). Subsequently, legs from each tercile were randomly assigned to either slow or fast tempo."
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	PY			See above
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N			Within-subjects design
Risk-of-bias judgment (low/high/some concerns)	Low risk			

Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		Y		It is a RT-study and within-subject, so yes.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	Y			Within-subject design
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN			Some dropped out, as is expected, but groups had a similar number of participants at the end, so it unlikely to have affected results.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?				
Risk-of-bias judgment (low/high/some concerns)		Low risk		
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?		N		≈28% of randomised participants dropped out due to adherence
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?			NI	No sensitivity analyses. Only the complete cases are of interest
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	PN			I see no reason to expect that dropouts were related to the outcome. Personal reasons were mentioned
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?				
Risk-of-bias judgment (low/high/some concerns)		Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?				
Bias in measurement of the outcome - STRENGTH				
4.1 Was the method of measuring the outcome inappropriate?	N			1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN			Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?			NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		Y		Encouragement, more attempts, etc.

4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Given the initial hypothesis and modest results, it is unlikely suspicion is needed that the assessment of the outcome was influenced by bias.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - HYPERTROPHY			
4.1 Was the method of measuring the outcome inappropriate?	PN		Ultrasound, although more prone to human error and judgement than MRI and CT, has been found valid as a measure of hypertrophy.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N		"The same 2 technicians (i.e., probe handling and screen measurement) performed all of the pretesting and post-testing measurements and were blinded to all limb assessments and the experimental condition assigned to each leg."
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?			
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?			
Risk-of-bias judgment (low/high/some concerns)	Low risk		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in selection of the reported result			
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:			
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?		NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?		NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?			
Overall bias			
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?			
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.			

**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Azevedo et al. (2022)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	No mention of methods, just randomised.
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	Same as above
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN			Within-subjects design
Risk-of-bias judgment (low/high/some concerns)	Some concerns			
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		Y		It is a RT-study and within-subject, so yes.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY			Steps were taken to account for nutrition and training outside of the intervention.
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	PN			Some dropped out, as is expected, but groups had a similar number of participants at the end, so it unlikely to have affected results.
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?				
Risk-of-bias judgment (low/high/some concerns)	Low risk			
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y			The study had no dropouts, and all measurements are reported
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?				
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?				
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?				
Risk-of-bias judgment (low/high/some concerns)	Low risk			

Optional: What is the predicted direction of bias due to missing outcome data?			
Bias in measurement of the outcome - STRENGTH			
4.1 Was the method of measuring the outcome inappropriate?	N		1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	Encouragement, more attempts, etc.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		My assessment is that the results and conclusions are relatively modest and I don't suspect any potential knowledge of the intervention influenced the outcome substantially.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in measurement of the outcome - HYPERTROPHY			
4.1 Was the method of measuring the outcome inappropriate?	PN		Ultrasound, although more prone to human error and judgement than MRI and CT, has been found valid as a measure of hypertrophy.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN		Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY	Given that ultrasound is inherently prone to human error and judgement, it could give rise to measurement error.
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN		Given the subject and that results contradicted their a priori hypotheses, I find it unlikely.
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?			
Bias in selection of the reported result			
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:			
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?		NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?		NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)	Some concerns		

Optional: What is the predicted direction bias due to selection of the reported results?				
Overall bias				
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?				
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.				
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*				

Segers et al. (2022)				
Bias domain and signalling questions	**Response options**			
Bias arising from the randomisation process	**Lower risk of bias**	**Higher risk of bias**	**Other**	**Justification for response**
1.1 Was the allocation sequence random?			NI	No mention of methods, just randomised and stratified by "starters" vs "non-starters".
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?			NI	Same as above
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	PN			Baseline looked similar.
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction of bias arising from the randomisation process?				
Bias due to deviations from intended interventions				
2.1 Were participants aware of their assigned intervention during the trial?		PY		It is a RT-study, so yes, probably. Unless they did not know the purpose of the study.
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		PY		No mention, but given the nature of the study, those supervising the training would be.
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?	PY			There is evidence that training was standardised during the experimental period. Though no mention of nutrition, the participants were elite football players, which can provide some assurances over other populations (e.g., more likely having access to professional nutrition practitioners).
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?	PN			None are apparent
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?	N			No drop outs
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to intervention?				

Risk-of-bias judgment (low/high/some concerns)		Low risk		
Optional: What is the predicted direction of bias due to deviations from intended interventions?				
Bias due to missing outcome data				
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y			The study had no dropouts, and all measurements are reported
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?				
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?				
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?				
Risk-of-bias judgment (low/high/some concerns)		Low risk		
Optional: What is the predicted direction of bias due to missing outcome data?				
Bias in measurement of the outcome - STRENGTH				
4.1 Was the method of measuring the outcome inappropriate?	N			1RM is the gold standard of externally valid strength measures.
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN			Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?			NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY		Encouragement, attempts etc
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN			No reason to suspect that belief or expectation influenced assessment. Hypothesis was not supported by the results.
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?				
Bias in measurement of the outcome - POWER				
4.1 Was the method of measuring the outcome inappropriate?	N			CMJ analysis via app seems appropriate
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	PN			Measurements were the same and performed at the same time for both groups.
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?			NI	No mention of blinding of assessors.
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		PY		Encouragement, attempts etc
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	PN			No reason to suspect that belief or expectation influenced assessment. Hypothesis was not supported by the results.
Risk-of-bias judgment (low/high/some concerns)		Some concerns		
Optional: What is the predicted direction of bias in measurement of the outcome?				
Bias in selection of the reported result				

5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?		NI	No mention of pre-specified protocols
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:			
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?		NI	No mention of pre-specified protocols
5.3 ... multiple eligible analyses of the data?		NI	No mention of pre-specified protocols
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the predicted direction bias due to selection of the reported results?			
Overall bias			
Risk-of-bias judgment (low/high/some concerns)	Some concerns		
Optional: What is the overall predicted direction of bias for this outcome?			
Y=yes; PY=probably yes; PN=probably no; N=no; NA=not applicable; NI=no information.			
**Signalling questions for bias due to deviations from intended interventions relate to the effect of assignment to intervention.*			